Gaussian Model

For cells (i, ℓ) with `model = gaussian`. In this section and the next four, the subscript $i\ell$ is dropped and k is the day.

Model. A mission increment is $\mathcal{N}(\mu_{jk}^{\text{inc}}, v_{jk}^{\text{inc}})$ (inputs `mu`, `v`); the initial damage is $\mathcal{N}(\mu_0, v_0)$; a replacement draw is $\mathcal{N}(\mu^{\text{new}}, v^{\text{new}})$. A known damage has variance 0. **Assumption N1.** The increments, the initial damage and the replacement draws are mutually independent.

Laws and tracked quantities. ARD_∞ and ARD_1. Tracked: μ_k and v_k; under ARD_1 also the anchors g_k^μ and g_k^v. Day maps:

$$\mu_k = \begin{cases} \mu_{k-1} + \delta\mu_k & \text{no intervention on day } k, \\ (1-\rho)\,\mu_{k-1} & \text{repair, } \text{ARD}_\infty, \\ (1-\rho)\,\mu_{k-1} + \rho\, g_{k-1}^\mu & \text{repair, } \text{ARD}_1, \\ \mu^{\text{new}} & \text{replacement}, \end{cases} \tag{1}$$

$$v_k = \begin{cases} v_{k-1} + \delta v_k & \text{no intervention on day } k, \\ (1-\rho)^2\, v_{k-1} & \text{repair, } \text{ARD}_\infty, \\ (1-\rho)^2\, v_{k-1} + \big(1-(1-\rho)^2\big)\, g_{k-1}^v & \text{repair, } \text{ARD}_1, \\ v^{\text{new}} & \text{replacement}, \end{cases} \tag{2}$$

with $\delta\mu_k = \sum_{j=1}^M x_{ijk}\,\mu_{jk}^{\text{inc}}$ and $\delta v_k = \sum_{j=1}^M x_{ijk}\, v_{jk}^{\text{inc}}$.

Big-M rows (`encoding = "bigm"`, the default). In this section and the next four, each row holds for $k = 1, \ldots, T$, and $e_k = x_k^m + x_k^r$. The ARD_∞ rows are built at ARD_∞ cells; the ARD_1 rows and the anchor rows at ARD_1 cells. With `allow_replacement = false`, $x_k^r = 0$ and the replacement rows are not built. Big-M constants: τ for μ and g^μ; $V_{\max} = \tau^2/\kappa_G^2$ for v and g^v, with $\kappa_G = \Phi^{-1}(1-\varepsilon)$.

Mean.

$$\text{no intervention:} \quad \mu_k \geq \mu_{k-1} + \delta\mu_k - \tau\,(x_k^m + x_k^r), \tag{3}$$

$$\text{repair, } \text{ARD}_\infty\text{:} \quad \mu_k \geq (1-\rho)\,\mu_{k-1} - \tau\,(1 - x_k^m), \tag{4}$$

$$\text{repair, } \text{ARD}_1\text{:} \quad \mu_k \geq (1-\rho)\,\mu_{k-1} + \rho\, g_{k-1}^\mu - \tau\,(1 - x_k^m), \tag{5}$$

$$\text{replacement:} \quad \mu_k \geq \mu^{\text{new}} - \tau\,(1 - x_k^r), \quad \mu_k \leq \mu^{\text{new}} + \tau\,(1 - x_k^r). \tag{6}$$

Variance.

$$\text{no intervention:} \quad v_k \geq v_{k-1} + \delta v_k - V_{\max}\,(x_k^m + x_k^r), \tag{7}$$

$$\text{repair, } \text{ARD}_\infty\text{:} \quad v_k \geq (1-\rho)^2\, v_{k-1} - V_{\max}\,(1 - x_k^m), \tag{8}$$

$$\text{repair, } \text{ARD}_1\text{:} \quad v_k \geq (1-\rho)^2\, v_{k-1} + \big(1-(1-\rho)^2\big)\, g_{k-1}^v - V_{\max}\,(1 - x_k^m), \tag{9}$$

$$\text{replacement:} \quad v_k \geq v^{\text{new}} - V_{\max}\,(1 - x_k^r), \quad v_k \leq v^{\text{new}} + V_{\max}\,(1 - x_k^r). \tag{10}$$

Anchors (ARD_1), for $(y, M_y) \in \{(\mu, \tau), (v, V_{\max})\}$:

$$g_k^y - y_k \leq M_y\,(1 - e_k), \qquad y_k - g_k^y \leq M_y\,(1 - e_k), \tag{11}$$

$$g_k^y - g_{k-1}^y \leq M_y\, e_k, \qquad g_{k-1}^y - g_k^y \leq M_y\, e_k. \tag{12}$$

Day 0 and replacement. The constants μ_0, v_0 and, under ARD_1, g_0^μ, g_0^v (Section **??**); the replacement values μ^{new}, v^{new} (Section **??**).

Indicator form (`encoding = "indicator"`): the rows Eqs. (**??**)–(**??**) with the same day maps, Eqs. (1)–(2).

Reliability row

Requires $\varepsilon < 0.5$ (Section **??**). On every day $k = 1, \dots, T$:

$$\kappa_G^2 \, v_k \le (\tau - \mu_k)^2, \qquad \mu_k \le \tau. \tag{13}$$

Forms. `exact`: Eq. (13) as written, a nonconvex quadratic constraint (`NonConvex` = 2, Section **??**). `tangent` and `pwl`: Section **??**, with $\chi = v$, $f(\mu) = (\tau - \mu)^2/\kappa_G^2$ and the side row $\mu \le \tau$.

Optional row. If `v_max_user` is given for the cell: $v_k \le V^{\text{user}}$ on every day.

Inverse Gaussian Model

For cells (i, ℓ) with `model` = `inverse_gaussian` (IG).

Model. $\text{IG}(\mu, \lambda)$ is the inverse Gaussian law with mean μ and shape λ. Every term of mean a (a mission increment, mean μ_{jk}^{inc}; the initial damage, μ_0; a replacement draw, μ^{new}) has the law $\text{IG}(a, a/\eta) = a\,Z$, with $Z \sim \text{IG}(1, 1/\eta)$ and $\eta > 0$ the dispersion of the cell (input `eta`). A known damage is a constant. No independence between terms is required (scoping document, section "Stochastic orderings").

Laws and tracked quantities. ARD_∞ and ARD_1. Tracked: μ_k; under ARD_1 also the anchor g_k^μ. Day map:

$$\mu_k = \begin{cases} \mu_{k-1} + \delta\mu_k & \text{no intervention on day } k, \\ (1-\rho)\,\mu_{k-1} & \text{repair, } \text{ARD}_\infty, \\ (1-\rho)\,\mu_{k-1} + \rho\, g_{k-1}^\mu & \text{repair, } \text{ARD}_1, \\ \mu^{\text{new}} & \text{replacement}, \end{cases} \tag{14}$$

with $\delta\mu_k = \sum_{j=1}^{M} x_{ijk}\, \mu_{jk}^{\text{inc}}$.

Big-M rows (`encoding = "bigm"`, the default; conditions in Section 0.1). Big-M constant: τ for μ and g^μ.

$$\text{no intervention:} \quad \mu_k \ge \mu_{k-1} + \delta\mu_k - \tau\,(x_k^m + x_k^r), \tag{15}$$

$$\text{repair, } \text{ARD}_\infty\text{:} \quad \mu_k \ge (1-\rho)\,\mu_{k-1} - \tau\,(1 - x_k^m), \tag{16}$$

$$\text{repair, } \text{ARD}_1\text{:} \quad \mu_k \ge (1-\rho)\,\mu_{k-1} + \rho\, g_{k-1}^\mu - \tau\,(1 - x_k^m), \tag{17}$$

$$\text{replacement:} \quad \mu_k \ge \mu^{\text{new}} - \tau\,(1 - x_k^r), \quad \mu_k \le \mu^{\text{new}} + \tau\,(1 - x_k^r). \tag{18}$$

Anchor (ARD_1):

$$g_k^\mu - \mu_k \le \tau\,(1 - e_k), \qquad \mu_k - g_k^\mu \le \tau\,(1 - e_k), \tag{19}$$

$$g_k^\mu - g_{k-1}^\mu \le \tau\, e_k, \qquad g_{k-1}^\mu - g_k^\mu \le \tau\, e_k. \tag{20}$$

Day 0 and replacement. The constants μ_0 and, under ARD_1, g_0^μ (Section **??**); the replacement value μ^{new} (Section **??**).

Indicator form (`encoding = "indicator"`): the rows Eqs. (**??**)–(**??**) with the same day map, Eq. (14).

Reliability row

Let q be the $(1-\varepsilon)$-quantile of Z, $\Pr(Z > q) = \varepsilon$, and $C = \mathbb{E}[Z \mid Z > q]$. On every day $k = 1, \dots, T$:

$$\mu_k \le \bar{\mu}, \qquad \bar{\mu} = \frac{\tau}{C}. \tag{21}$$

The row is linear: it is built in its exact form, and a value of `reliability_impl` at the cell is ignored.

Computation of $\bar{\mu}$ (offline, once per cell). With $r(x) = 1/\sqrt{\eta x}$, the distribution function of Z is

$$F(x) = \Phi\big(r(x)\,(x-1)\big) + e^{2/\eta}\,\Phi\big(-r(x)\,(x+1)\big).$$

The second term is computed as $\exp\big(2/\eta + \ln\Phi(-r(x)(x+1))\big)$, with $\ln\Phi$ from a log-space routine (`scipy.special.log_ndtr`).

1. q: the root of $1 - F(q) = \varepsilon$, by bisection or `brentq` on $\ln q$; equivalently, the method `isf` of `scipy.stats.invgauss` with `mu` $= \eta$, `scale` $= 1/\eta$.
2. The tail mean: $\mathbb{E}[Z;\, Z > q] = F(1/q)$, so $\bar{\mu} = \tau\,\varepsilon / F(1/q)$. Quadrature of $\int_q^\infty z\, f_Z(z)\, dz$, with $f_Z(z) = (2\pi\eta z^3)^{-1/2} \exp\big(-(z-1)^2/(2\eta z)\big)$, gives the same value.

Wiener Model

For cells (i, ℓ) with `model` = `wiener`.

Model. A mission day adds $\mathcal{N}(\mu_{jk}^{\text{inc}}, \sigma^2)$, with the mean from `mu` and one diffusion constant $\sigma > 0$ per cell (input `sigma`); the initial damage is $\mathcal{N}(\mu_0, v_0)$, $v_0 \ge 0$; a replacement draw is $\mathcal{N}(\mu^{\text{new}}, v^{\text{new}})$. **Assumption W1.** These terms are mutually independent. A repair acts on the mean only; the variance falls only at a replacement.

Laws and tracked quantities. ARD_∞ and ARD_1. Tracked: μ_k and v_k; under ARD_1 also the mean anchor g_k^μ (no variance anchor). The mean follows Eq. (1). The variance follows

$$v_k = \begin{cases} v_{k-1} + \delta v_k & \text{no intervention on day } k, \\ v_{k-1} & \text{repair, both laws}, \\ v^{\text{new}} & \text{replacement}, \end{cases} \qquad \delta v_k = \sigma^2 \sum_{j=1}^{M} x_{ijk}. \tag{22}$$

Big-M rows (`encoding = "bigm"`, the default; conditions in Section 0.1). Big-M constants: τ for μ and g^μ; $V_{\max} = \tau^2/\kappa_G^2$ for v, with $\kappa_G = \Phi^{-1}(1-\varepsilon)$.

Mean.

$$\text{no intervention:} \quad \mu_k \ge \mu_{k-1} + \delta\mu_k - \tau\,(x_k^m + x_k^r), \tag{23}$$

$$\text{repair, } \text{ARD}_\infty\text{:} \quad \mu_k \ge (1-\rho)\,\mu_{k-1} - \tau\,(1 - x_k^m), \tag{24}$$

$$\text{repair, } \text{ARD}_1\text{:} \quad \mu_k \ge (1-\rho)\,\mu_{k-1} + \rho\, g_{k-1}^\mu - \tau\,(1 - x_k^m), \tag{25}$$

$$\text{replacement:} \quad \mu_k \ge \mu^{\text{new}} - \tau\,(1 - x_k^r), \quad \mu_k \le \mu^{\text{new}} + \tau\,(1 - x_k^r). \tag{26}$$

Variance (both laws; no anchor).

$$\text{no intervention, repair:} \quad v_k \ge v_{k-1} + \delta v_k - V_{\max}\, x_k^r, \tag{27}$$

$$\text{replacement:} \quad v_k \ge v^{\text{new}} - V_{\max}\,(1 - x_k^r), \quad v_k \le v^{\text{new}} + V_{\max}\,(1 - x_k^r). \tag{28}$$

Anchor (ARD_1, mean only):

$$g_k^\mu - \mu_k \le \tau\,(1 - e_k), \qquad \mu_k - g_k^\mu \le \tau\,(1 - e_k), \tag{29}$$

$$g_k^\mu - g_{k-1}^\mu \le \tau\, e_k, \qquad g_{k-1}^\mu - g_k^\mu \le \tau\, e_k. \tag{30}$$

Day 0 and replacement. The constants μ_0, v_0 and, under ARD_1, g_0^μ (Section **??**); the replacement values μ^{new}, v^{new} (Section **??**).

Indicator form (`encoding = "indicator"`): the rows Eqs. (**??**)–(**??**) with the same day maps, Eqs. (1) and (22): the variance has $\psi = v_{k-1}$, and only g^μ has anchor rows.

Reliability row. The Gaussian row, Eq. (13), with $\kappa_G = \Phi^{-1}(1-\varepsilon)$, $\varepsilon < 0.5$, $V_{\max} = \tau^2/\kappa_G^2$, on every day $k = 1, \ldots, T$, in the same forms and with the same f and side row (Section 0.1). The optional row $v_k \le V^{\mathrm{user}}$ is added when `v_max_user` is given.

Note. The loop row $v_T \le v_{H_1}$ forces a replacement of the component in the operating phase whenever its train flies a mission there; with `allow_replacement` false, the train cannot fly in the operating phase (scoping document, section "Stochastic orderings", loop constraint for each degradation model).

Gamma Model

For cells (i, ℓ) with `model` = `gamma`.

Model. $\mathrm{Gamma}(a, \beta)$ has shape a, rate β and mean a/β. Terms:

- a mission increment with mean $\mu_{jk}^{\mathrm{inc}} > 0$ (`mu`) and rate β_{jk} (`gamma_beta`, `gamma_beta_trans`): $\mathrm{Gamma}(\beta_{jk}\mu_{jk}^{\mathrm{inc}}, \beta_{jk})$;
- the initial damage, $\mu_0 > 0$: $\mathrm{Gamma}(\beta_0\mu_0, \beta_0)$ when `gamma_beta_0` $= \beta_0$ is given; the known constant μ_0 when it is omitted;
- a replacement draw, $\mu^{\mathrm{new}} > 0$: $\mathrm{Gamma}(\beta^{\mathrm{new}}\mu^{\mathrm{new}}, \beta^{\mathrm{new}})$ when `gamma_beta_new` $= \beta^{\mathrm{new}}$ is given; the known constant μ^{new} when it is omitted.

Assumption G1. All these terms are mutually independent.

Laws and tracked quantities. ARD_∞ and ARD_1. Tracked: μ_k (the shape is not tracked); under ARD_1 also the anchor g_k^μ. Day map:

$$\mu_k = \begin{cases} \mu_{k-1} + \delta\mu_k & \text{no intervention on day } k, \\ (1-\rho)\,\mu_{k-1} & \text{repair, } \mathrm{ARD}_\infty, \\ (1-\rho)\,\mu_{k-1} + \rho\, g_{k-1}^\mu & \text{repair, } \mathrm{ARD}_1, \\ \mu^{\mathrm{new}} & \text{replacement,} \end{cases} \tag{31}$$

with $\delta\mu_k = \sum_{j=1}^{M} x_{ijk}\,\mu_{jk}^{\mathrm{inc}}$.

Big-M rows (`encoding = "bigm"`, the default; conditions in Section 0.1). Big-M constant: τ for μ and g^μ.

$$\text{no intervention:} \quad \mu_k \ge \mu_{k-1} + \delta\mu_k - \tau\,(x_k^m + x_k^r), \tag{32}$$

$$\text{repair, } \mathrm{ARD}_\infty\text{:} \quad \mu_k \ge (1-\rho)\,\mu_{k-1} - \tau\,(1 - x_k^m), \tag{33}$$

$$\text{repair, } \mathrm{ARD}_1\text{:} \quad \mu_k \ge (1-\rho)\,\mu_{k-1} + \rho\, g_{k-1}^\mu - \tau\,(1 - x_k^m), \tag{34}$$

$$\text{replacement:} \quad \mu_k \ge \mu^{\mathrm{new}} - \tau\,(1 - x_k^r), \quad \mu_k \le \mu^{\mathrm{new}} + \tau\,(1 - x_k^r). \tag{35}$$

Anchor (ARD_1):

$$g_k^\mu - \mu_k \le \tau\,(1 - e_k), \qquad \mu_k - g_k^\mu \le \tau\,(1 - e_k), \tag{36}$$

$$g_k^\mu - g_{k-1}^\mu \le \tau\, e_k, \qquad g_{k-1}^\mu - g_k^\mu \le \tau\, e_k. \tag{37}$$

Day 0 and replacement. The constants μ_0 and, under ARD_1, g_0^μ (Section **??**); the replacement value μ^{new} (Section **??**).

Indicator form (`encoding = "indicator"`): the rows Eqs. (**??**)–(**??**) with the same day map, Eq. (31).

Smallest rate.

$$\beta^* = \min\Big(\{\beta_{jk} : \mu^{\text{inc}}_{jk} > 0\} \cup \{\beta_0 : \mu_0 > 0\} \cup \{\beta^{\text{new}} : \mu^{\text{new}} > 0\}\Big), \tag{38}$$

over the missions j and the days k of both phases. β_0 and β^{new} enter only when given, and β^{new} only when `allow_replacement` is true. A cell in which no term has a rate has the row $\mu_k \le \tau$ in place of Eq. (39).

Reliability row

$Q(a, x) = \Pr(\text{Gamma}(a, 1) > x)$ is the regularised upper incomplete gamma function (`gammaincc` of `scipy.special`). Let $t = \beta^*\tau$ and, for $A \in [0, t)$,

$$w(A) = \max_{c \in (0,1]} Q\left(\frac{A}{c}, \frac{t}{c}\right).$$

The sharp cap $A_1(t, \varepsilon)$ is the largest $A \in [0, t)$ with $w(A) \le \varepsilon$. For $\varepsilon \ge 1/2$ every $A < t$ satisfies this, and A_1 is the value that step 3 below returns, which lies below t. On every day $k = 1, \dots, T$:

$$\mu_k \le \frac{A_1(\beta^*\tau,\, \varepsilon)}{\beta^*}. \tag{39}$$

The row is linear: it is built in its exact form, and a value of `reliability_impl` at the cell is ignored.

Computation of A_1 (offline, once per cell):

1. $\hat{A}$: the root of $Q(\hat{A}, t) = \varepsilon$, by bisection on $\hat{A}$ (`gammaincinv` solves for the second argument and does not apply).
2. $w(A)$: maximum of $c \mapsto Q(A/c, t/c)$ over a logarithmic grid of at least 400 values of c in $[\min(10^{-4}, t/100), 1]$, refined by a golden-section search between the neighbours of the best grid point. If the best grid point is the smallest c, the grid is extended downwards.
3. If $\hat{A} < t$ and $w(\hat{A}) \le \varepsilon$: $A_1 = \hat{A}$. Otherwise: bisection on A in $[\varepsilon t, \min(\hat{A}, t))$, to a relative tolerance of 10^{-9}; $w(\varepsilon t) \le \varepsilon$ holds. The bisection returns the lower end of its last interval.

Rainflow Model

For cells (i, ℓ) with `model` = `rainflow`. The law of a mission increment comes from rainflow counting of recorded load histories (Palmgren–Miner damage) and enters only through summary statistics: mean, variance, width of its range, or cumulant generating function (CGF) at a fixed s. The tail bound is chosen per cell by `bound_method`: `markov`, `cantelli`, `hoeffding`, `bernstein` or `chernoff`.

Assumptions on the input data.

- **A0.** Every increment, the initial damage and every replacement draw are non-negative.
- **A1.** The mission increments, the initial damage and the replacement draws are mutually independent (all bounds except Markov).
- **A2** (Hoeffding, Bernstein). Each increment lies in an interval of width b^{inc}_{jk} (input `support`). The initial damage is a known part plus independent terms in intervals, with sum of squared widths R_0 (`R_0`) and largest width b^0 (`support_0`); a replacement draw likewise, with R^{new}, b^{new} (`R_new`, `support_new`). A known damage has $R = b = 0$, the defaults.
- **A3.** After day T the operating phase repeats with period H_2 (Section **??**).

The *width* of X is the width b of an interval $[l, l+b]$ that contains X almost surely; it must satisfy $v \le b^2/4$ (Section **??**).

Tracked quantities per bound. Every cell tracks μ; the bound adds at most one quantity; under ARD_1 each tracked quantity has an anchor (g^μ and g^v, g^R or g^K). Both laws are accepted for every bound. "Forms" lists the forms in which the row is built ("all three": `exact`, `tangent`, `pwl`); at a linear row a value of `reliability_impl` is ignored.

`bound_method`	Tracked	Increment inputs	Initial; replacement inputs	Forms
`markov`	μ	`mu`	–	`exact`
`cantelli`	μ, v	`mu`, `v`	`v_0`; `replacement_v`	all three
`hoeffding`	μ, R	`mu`, `support`	`R_0`; `R_new`	all three
`bernstein`	μ, v	`mu`, `v`, `support`	`v_0`, `support_0`; `replacement_v`, `support_new`	all three
`chernoff`	μ, K	`mu`, `cgf`, `s_chernoff`	`cgf_0`; `cgf_new`	`exact`

R is the Hoeffding budget (sum of the squared widths of the terms of the damage). K is the CGF of the damage at $s =$ `s_chernoff` > 0. Day increments: $\delta\mu_k = \sum_{j=1}^{M} x_{ijk}\, \mu_{jk}^{\mathrm{inc}}$, $\delta v_k = \sum_j x_{ijk}\, v_{jk}^{\mathrm{inc}}$, $\delta R_k = \sum_j x_{ijk}\, (b_{jk}^{\mathrm{inc}})^2$, $\delta K_k = \sum_j x_{ijk}\, K_{jk}^{\mathrm{inc}}$, with $K_{jk}^{\mathrm{inc}} = \ln \mathbb{E}\, e^{sX_{jk}}$ (input `cgf`).

Day maps. Each tracked quantity follows the generic map of Section **??** with its factor a: $1-\rho$ for μ and K, $(1-\rho)^2$ for v and R. It adds its day increment on a day without intervention; a repair multiplies it by a under ARD_∞, and gives a times its value plus $1-a$ times its anchor under ARD_1; a replacement sets it to its replacement value. Its anchor is its value just after the last intervention.

Starting and replacement values. μ and v: from the inputs (Section **??**). R_0 from `R_0` and R^{new} from `R_new`, default 0. K_0 from `cgf_0`, default $s\,\mu_0$; K^{new} from `cgf_new`, default $s\,\mu^{\mathrm{new}}$. With `replacement_as_new`, all replacement values are 0. The defaults describe a known damage; a random initial or replacement damage needs its own `R_0`, `cgf_0`, `R_new`, `cgf_new` (scoping document, section "Conservative surrogates for the empirical damage-increment distribution").

Big-M rows (`encoding = "bigm"`, the default; conditions in Section 0.1), with $\kappa = \ln(1/\varepsilon)$ and c from Eq. (61). The rows of μ are built at every cell, the rows of v at Cantelli and Bernstein cells, the rows of R at Hoeffding cells, and the rows of K at Chernoff cells. Big-M constants: τ for μ and g^μ; $M_v = V^{\mathrm{C}} = \varepsilon\tau^2/(1-\varepsilon)$ (Cantelli) or $M_v = V^{\mathrm{B}} = \tau(\tau - 2c)/(2\kappa)$ (Bernstein) for v and g^v; $2\tau^2/\kappa$ for R and g^R; $s\tau - \kappa$ for K and g^K.

Mean (every bound).

$$\text{no intervention:} \quad \mu_k \ge \mu_{k-1} + \delta\mu_k - \tau\,(x_k^m + x_k^r), \tag{40}$$

$$\text{repair, } \mathrm{ARD}_\infty\text{:} \quad \mu_k \ge (1-\rho)\,\mu_{k-1} - \tau\,(1 - x_k^m), \tag{41}$$

$$\text{repair, } \mathrm{ARD}_1\text{:} \quad \mu_k \ge (1-\rho)\,\mu_{k-1} + \rho\, g_{k-1}^{\mu} - \tau\,(1 - x_k^m), \tag{42}$$

$$\text{replacement:} \quad \mu_k \ge \mu^{\mathrm{new}} - \tau\,(1 - x_k^r), \quad \mu_k \le \mu^{\mathrm{new}} + \tau\,(1 - x_k^r). \tag{43}$$

Variance (Cantelli, Bernstein).

$$\text{no intervention:} \quad v_k \ge v_{k-1} + \delta v_k - M_v\,(x_k^m + x_k^r), \tag{44}$$

$$\text{repair, } \mathrm{ARD}_\infty\text{:} \quad v_k \ge (1-\rho)^2\, v_{k-1} - M_v\,(1 - x_k^m), \tag{45}$$

$$\text{repair, } \mathrm{ARD}_1\text{:} \quad v_k \ge (1-\rho)^2\, v_{k-1} + \big(1 - (1-\rho)^2\big)\, g_{k-1}^{v} - M_v\,(1 - x_k^m), \tag{46}$$

$$\text{replacement:} \quad v_k \ge v^{\mathrm{new}} - M_v\,(1 - x_k^r), \quad v_k \le v^{\mathrm{new}} + M_v\,(1 - x_k^r). \tag{47}$$

Hoeffding budget (Hoeffding).

$$\text{no intervention:} \quad R_k \geq R_{k-1} + \delta R_k - (2\tau^2/\kappa)\,(x_k^m + x_k^r), \tag{48}$$
$$\text{repair, ARD}_\infty\text{:} \quad R_k \geq (1-\rho)^2\, R_{k-1} - (2\tau^2/\kappa)\,(1 - x_k^m), \tag{49}$$
$$\text{repair, ARD}_1\text{:} \quad R_k \geq (1-\rho)^2\, R_{k-1} + \big(1-(1-\rho)^2\big)\, g_{k-1}^R - (2\tau^2/\kappa)\,(1 - x_k^m), \tag{50}$$
$$\text{replacement:} \quad R_k \geq R^{\text{new}} - (2\tau^2/\kappa)\,(1 - x_k^r), \quad R_k \leq R^{\text{new}} + (2\tau^2/\kappa)\,(1 - x_k^r). \tag{51}$$

Chernoff quantity (Chernoff).

$$\text{no intervention:} \quad K_k \geq K_{k-1} + \delta K_k - (s\tau - \kappa)\,(x_k^m + x_k^r), \tag{52}$$
$$\text{repair, ARD}_\infty\text{:} \quad K_k \geq (1-\rho)\, K_{k-1} - (s\tau - \kappa)\,(1 - x_k^m), \tag{53}$$
$$\text{repair, ARD}_1\text{:} \quad K_k \geq (1-\rho)\, K_{k-1} + \rho\, g_{k-1}^K - (s\tau - \kappa)\,(1 - x_k^m), \tag{54}$$
$$\text{replacement:} \quad K_k \geq K^{\text{new}} - (s\tau - \kappa)\,(1 - x_k^r), \quad K_k \leq K^{\text{new}} + (s\tau - \kappa)\,(1 - x_k^r). \tag{55}$$

Anchors (ARD$_1$), for each pair $(y, M_y) \in \{(\mu, \tau),\ (v, M_v),\ (R, 2\tau^2/\kappa),\ (K, s\tau - \kappa)\}$ whose quantity y the cell tracks:

$$g_k^y - y_k \leq M_y\,(1 - e_k), \qquad y_k - g_k^y \leq M_y\,(1 - e_k), \tag{56}$$
$$g_k^y - g_{k-1}^y \leq M_y\, e_k, \qquad g_{k-1}^y - g_k^y \leq M_y\, e_k. \tag{57}$$

Day 0 and replacement. For the quantities that the cell tracks: the constants μ_0, v_0, R_0, K_0 and, under ARD$_1$, g_0^μ, g_0^v, g_0^R, g_0^K (Section **??**); the replacement values μ^{new}, v^{new}, R^{new}, K^{new} (paragraph "Starting and replacement values").

Indicator form (`encoding = "indicator"`): the rows Eqs. (**??**)–(**??**) with the same day maps (paragraph "Day maps").

Reliability rows

Each row holds on every day $k = 1, \ldots, T$.

Markov (`markov`):

$$\mu_k \leq \varepsilon\,\tau. \tag{58}$$

Cantelli (`cantelli`), with $\kappa_C = \sqrt{(1-\varepsilon)/\varepsilon}$:

$$\kappa_C^2\, v_k \leq (\tau - \mu_k)^2, \qquad \mu_k \leq \tau. \tag{59}$$

Hoeffding (`hoeffding`):

$$(\tau - \mu_k)^2 \geq \frac{\kappa}{2}\, R_k, \qquad \mu_k \leq \tau. \tag{60}$$

Bernstein (`bernstein`). Support constant, precomputed per cell:

$$b = \max\Big(b^0,\ b^{\text{new}},\ \max_{j,k}\, b_{jk}^{\text{inc}}\Big), \qquad c = \frac{\kappa\, b}{3}, \tag{61}$$

over the missions j and the days k of both phases; b^{new} enters only when `allow_replacement` is true. Row, with its side row:

$$(\tau - c - \mu_k)^2 \geq c^2 + 2\kappa\, v_k, \qquad \mu_k \leq \tau - c. \tag{62}$$

Requires $\tau > 2\kappa b/3$ (Section **??**).

Chernoff (`chernoff`), with $s =$ `s_chernoff`:

$$K_k - s\,\tau \le \ln \varepsilon. \tag{63}$$

Requires $s > \kappa/\tau$ and $K^{\text{new}} \le s\tau - \kappa$ (Section **??**).

Forms. Markov and Chernoff rows are linear and built in their exact form. The Cantelli, Hoeffding and Bernstein rows take the forms of Section **??** with:

Bound	χ	$f(\mu)$	Side row	$f(0)$
Cantelli	v	$\varepsilon(\tau-\mu)^2/(1-\varepsilon)$	$\mu \le \tau$	$\varepsilon\tau^2/(1-\varepsilon)$
Hoeffding	R	$2(\tau-\mu)^2/\kappa$	$\mu \le \tau$	$2\tau^2/\kappa$
Bernstein	v	$\big((\tau-\mu)^2 - 2c(\tau-\mu)\big)/(2\kappa)$	$\mu \le \tau - c$	$\tau(\tau-2c)/(2\kappa)$

At a Bernstein cell with $b > 0$ the tangent point must satisfy $\mu_r < \sqrt{\tau(\tau-2c)}$, and the `pwl` form requires $\tau/(2P) < \sqrt{\tau(\tau-2c)}$, $P =$ `pwl_points` (Section **??**).

Optional row. If `v_max_user` is given for a Cantelli or Bernstein cell: $v_k \le V^{\text{user}}$ on every day.

References

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